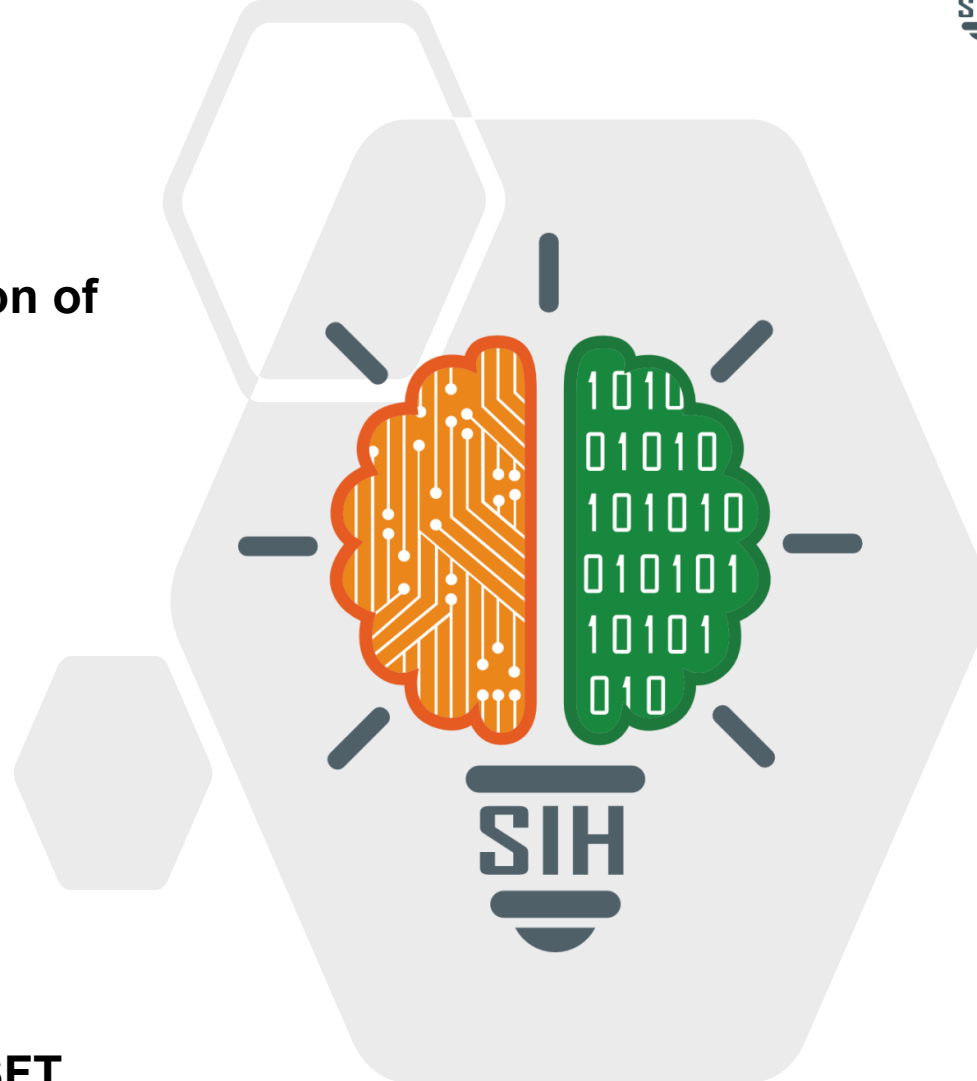


SMART INDIA HACKATHON 2026



- **Problem Statement ID : SIH26145**
- **Problem Statement Title : AI-Based Detection of Threats in Unidirectional IP Traffic**
- **Theme : Blockchain & Cybersecurity**
- **PS Category : Software**
- **Team ID-153388**
- **Team Name - EigenThinkers**

**TEAM TRACK RECORD : FITT, IIT DELHI
INCUBATION SUPPORT
(CREDITS : TEAM LEADER) : SAMSUNG SFT
2025 TOP 20 GRAND FINALE, IRIS NATIONAL
FAIR 2024 , FITT NAVACHAR MANTRA ONGOING**



The Problem

Failure of TRADITIONAL IDS Systems in critical infrastructures where traffic reaches the monitoring enclave through a data diode. The monitor gets a one-way copy: no replies, no handshakes, no way to probe or block. Current IDS expects a two-way link and known signatures. Therefore :

- Blind to offline zero-days & Encrypted TLS C2 CHANNELS
- Such Airgapped environments make Static signature database based mechanism obsolete.

```
16952 45.916978 92.223.116.197 10.35.93.179 TCP 2594 80 → 36486 [PSH, ACK] Seq=7444770 A
16953 45.917108 10.35.93.179 92.223.116.197 TCP 66 36486 → 80 [ACK] Seq=3116 Ack=74472
16954 45.920089 92.223.116.197 10.35.93.179 TCP 2594 80 → 36486 [PSH, ACK] Seq=7447298 A
16955 45.920089 2a03:20c1:7121::2 2409:4000:2400:9000::131 TCP 1369 80 → 36485 [ACK] Seq=228375 Ack=1 W
16956 45.926186 92.223.116.197 10.35.93.179 TCP 2594 80 → 36486 [PSH, ACK] Seq=7449826 A
```

```
> Internet Protocol Version 6, Src: 2a03:20c1:7121::2, Dst: 2409:4000:2400:9000::131
Transmission Control Protocol, Src Port: 80, Dst Port: 53988, Seq: 192451, Ack: 1, Len: 1283
  Source Port: 80
  Destination Port: 53988
  [Stream index: 8]
  [Conversation completeness: Incomplete (12)]
  [TCP Segment Len: 1283]
  Sequence Number: 192451 (relative sequence number)
  Sequence Number (raw): 3478708778
  [Next Sequence Number: 193734 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 3304618860
  1000 .... = Header Length: 32 bytes (8)
  [Flags: 0x010 (ACK)]
  Window: 8
  [Calculated window size: 8]
  [Window size scaling factor: -1 (unknown)]
  Checksum: 0xe627 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
Options: (12 bytes). No-Operation (NOP). No-Operation (NOP). Timestamps
```

```
0030
0040
0050
0060
0070
0080
0090
00a0
00b0
00c0
00d0
00e0
00f0
0100
0110
0120
0130
0140
0150
0160
0170
0180
0190
01a0
01b0
01c0
```

Our Solution

- UniAethel is a passive behavioural engine that works only on one-way traffic metadata. Four detection engines that each monitor different aspects of the network traffic and produce multidimensional anomaly scores . A separate layer combines them into a threat state to predict attack vector & trajectory, so a real attack stands out and a harmless spike does not.

Target Attack Vectors

- Encrypted Traffic Threat Vectors
- Offline Air-Gap Bypass Vectors : Physical Malware Implants . e.g. Stuxnet etc
- C2 Beaconing , DDoS, Reconnaissance, Data Exfiltration

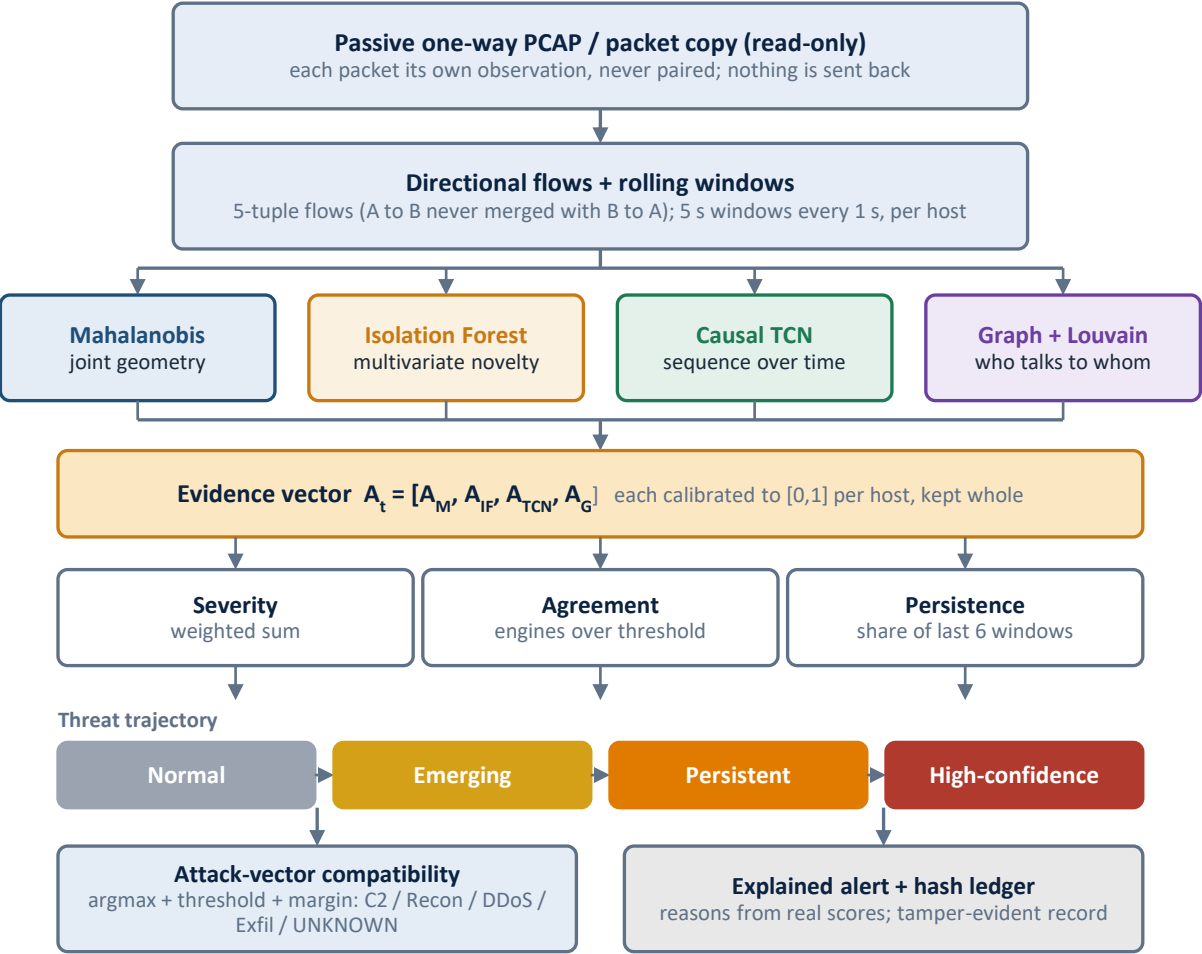
Core Novelties

- Four independent views of the same one-way traffic: distribution, multivariate novelty, time, and who talks to whom. Only one KIND OF ANOMALY is not considered an alert.
- Each host is judged against its own normal, not the network average.
- An explicit attack-vector layer classifies C2, recon, DDoS or exfiltration only when the evidence clearly fits one profile. Otherwise it says UNKNOWN instead of guessing.
- Evidence is never squeezed into one score. Severity, how many engines agree, how long it lasts and the threat state are kept apart.
- Every alert explains itself and is written to a tamper-evident hash chain.
- Tested honestly: weights and thresholds were locked and hashed before the test data existed.

TECHNICAL APPROACH



System architecture



Only persistent, corroborated incidents are classified. The label is given only when one profile clearly wins; otherwise UNKNOWN.

What each engine answers, and what removing it costs

Engine	Question it asks	Measured evidence
Mahalanobis (Ledoit-Wolf)	Is the joint pattern of volume and timing off for this host? SUITS BEST DUE TO CORRELATION BW TIMING AND IP	Without it: 16 of 16 drops to 8
Isolation Forest	Is this a combination normal traffic never visits?	Without it: still 16 (no gain here, stated)
Causal TCN	Is the host's sequence departing from what its past predicts?	Without it: 16 drops to 12 (fewer false alerts)
Graph + Louvain	New peers, recurring, crossing communities?	Without it: 16 of 16 drops to 4

Key MODEL

Mahalanobis: $D^2 = (x-\mu)^T \Sigma^{-1} (x-\mu)$, Σ by Ledoit-Wolf shrinkage because traffic features are collinear.
Calibration: per host, on normal validation data only. $A = 0.99$ means “as rare as the top 1% of this host's normal”.
Graph: directed rolling graph (30 s); undirected copy only for Louvain. Score = max over signals of $-\log_{10} P(\text{signal} \geq x \mid \text{normal})$.
Severity: $S = 1 - \prod (1-A_i)^{w_i}$, equal w_i (linear sum tested: 4 of 16). $C = \sum 1[A_i \geq 0.99]$. P = share of last 6 windows with $S > \tau$.
Attack vector: $S_k = (C/4) \cdot P \cdot \sum w_{kj} \phi_{kj}$; label = argmax only if $\max \geq \tau_{\text{class}}$ and margin $\geq \delta$, else UNKNOWN.
Evidence: USING PBFT CONSENSUS INSTEAD OF PoW for Low End Hardware optimization

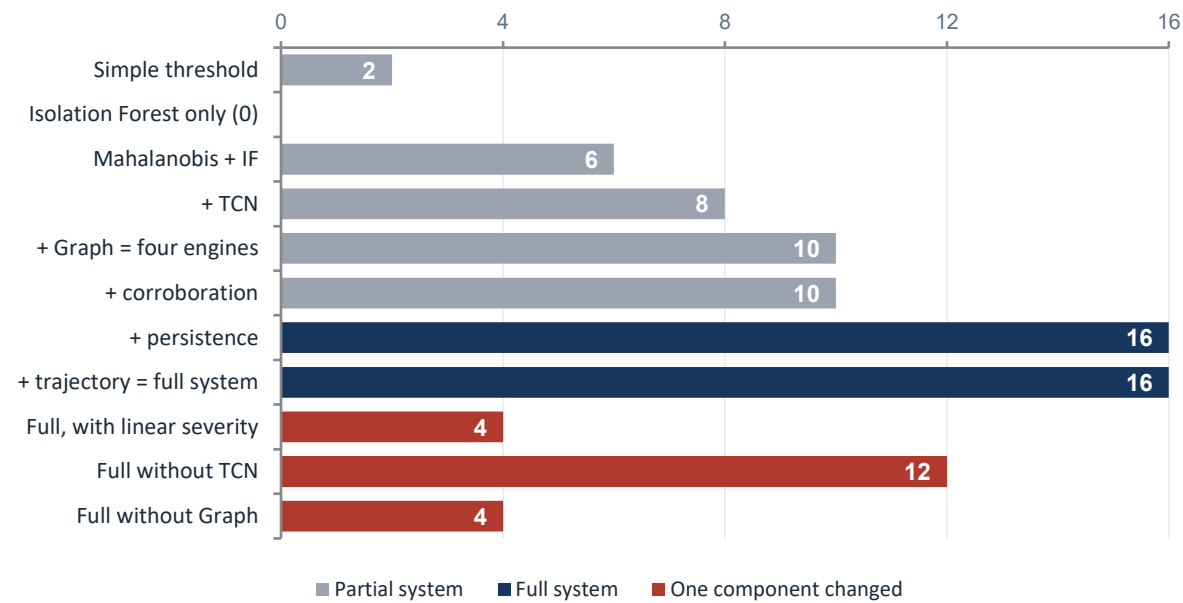
Stack

Scapy, dpkt, NumPy, scikit-learn, NetworkX, FastAPI. TCN written from scratch in NumPy, gradient-checked. 38 automated tests, including one-way (reverse traffic never needed), TCN causality and ledger tampering. No GPU, no payload decryption. PLANNING TO IMPLEMENT SNN FOR C2 BEACONING DETECTION IN FUTURE

FEASIBILITY AND VIABILITY



Working prototype: unseen attacks detected (of 16), by design choice



Every layer is tested: persistence lifts detection from 10 to 16; removing the graph drops it to 4; a linear severity sum manages only 4.

Held-out results (28 unseen PCAP replays, 2.3 h)

Measure	Result
Unseen attacks detected (full system)	16 / 16, median 19 s
Attack types confused with each other	0
Known attacks named correctly	8 / 14 (other 6 left UNKNOWN)
Never-seen attack left as UNKNOWN	5 / 5 incidents
C2 named correctly, jitter 0 to 50%	10 / 10
Detected at 50% packet loss	4 / 4 attacks
False alerts on normal traffic	9.3 per hour (main weakness)
Throughput on one CPU core	22,011 packets/s, 162 Mbps
Tamper tests on alert log	every edit and deletion caught
Automated tests	38 passing

Synthetic PCAPs of a simulated 22-host network, one-way view only. Thresholds hashed before the test set was scored; test scored once.

DRAWBACKS OF THE RECENT VERSION, and the fix

Weak spot	Why	Next step
Too many false alerts	Host baselines learned from only 2 benign captures	More benign data, calibrate apart from tuning
Recon labelled UNKNOWN	Recon profile scores too weak for its evidence	Re-weight the recon profile on validation
Benign periodic agents called C2	Behaviour alone can't tell a new heartbeat from a beacon	Known-service allow-list; label as a hint
1 MB exfiltration missed	Inside this host's normal transfer range	Longer horizon, cumulative volume

Built and tested: Scapy PCAP generator (14 scenario types + 10 benign look-alikes), directional flows, four engines, attack-vector layer with UNKNOWN rejection, explained incidents, hash ledger, API, dashboard.

Next: CIC-IDS2017 / CTU-13 replay, live streaming, egress-only baselines, diode testbed.

Reproduce: python scripts/run_all.py (about 10 minutes, deterministic seeds).

Evidence on file: frozen thresholds, dataset manifest, ablation, robustness, benchmark, 11 docs.

IMPACT AND BENEFITS



Who would use it

NTRO and defence enclaves: monitoring behind hardware data diodes

Critical infrastructure: SCADA segments in power, rail and refineries

SOC and CERT teams: alerts that come with evidence, not just a label

Forensic investigators: an alert trail that can't be edited later

What changes for them

Operations

Analysts get a named attack vector only when the evidence clearly fits, plus four scores and the reasons. Ambiguous cases are marked UNKNOWN, not mislabelled.

Security

New attack patterns are caught without waiting for signature updates to cross the air gap.

Cost

One ordinary CPU core processes 22,000 packets per second (162 Mbps) on an open-source stack. No GPU and no per-sensor licence.

0

packets sent into the protected network

16/16

unseen attacks detected, median 19 s

5/5

never-seen attack left as UNKNOWN, not guessed

162

Mbps processed on one CPU core

What a one-way link takes away, and how we cope

Limitation of a one-way feed	How UniAethel handles it
No return path	Fully passive: reads a copy, never sends a packet
Can't probe or verify a host	Judges each host against its own learned normal
Payload is encrypted	Uses sizes, timing, destinations and who talks to whom
Replies and handshakes may be missing	Marks unseen data as missing, never as zero
Attack never seen before	Flags it and says UNKNOWN instead of guessing (5 of 5)
Any single sign is weak	Four engines; agreement and persistence set confidence
Only the current moment is visible	Threat state tracks how behaviour is evolving
Logs could be altered	Data logged into decentralized ledger , USING PBFT IN PLACE OF PoW mechanism to run on low end hardware

RESEARCH AND REFERENCES

Inspired by Previous Research Manuscript |
2024-2026 (SAMEER SAAWAN)
“A Novel Blockchain Based Energy Anti-Theft
System Advanced with Machine Learning
Algorithm for Real-Time Anomaly Detection”



ADAPTION

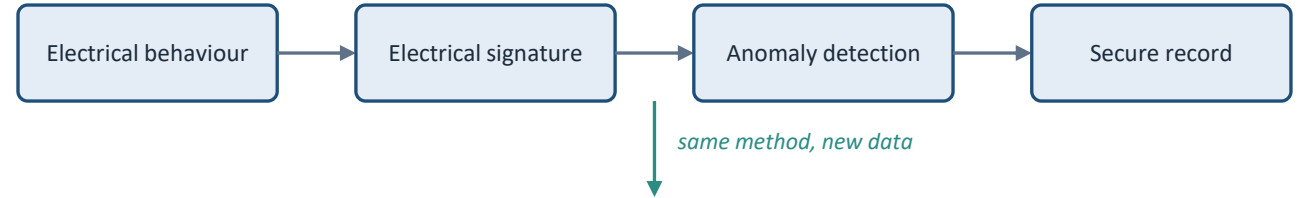
External recognition

Samsung Solve for Tomorrow 2025: Top 20 finalist, ₹1 lakh grant
FITT, IIT Delhi (2025): external innovation validation, Current shortlist at stage 1 Navachar Mantra
Samsung R&D (2025): technical evaluation
IRIS National Fair (2024): research recognition

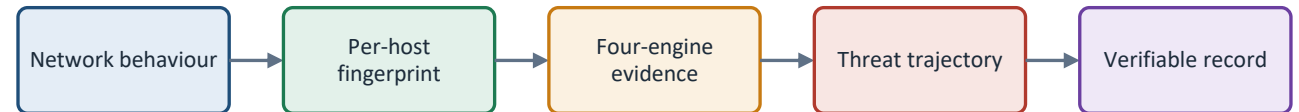


From power-theft detection to one-way network defence

Earlier work: power grid



UniAethel: one-way network traffic



References

PREVIOUS MANUSCRIPT (CryptEthEra, smart-grid theft detection):
https://github.com/SAMEER-MLPROJECT/AI_ANTI-THEFT_SYSTEM

PREVIOUS RESEARCH PAPER 2024-26
https://github.com/SAMEER-MLPROJECT/AI_ANTI-THEFT_SYSTEM/blob/main/research/research_paper_final.pdf

SIH26145 , (CURRENT) 2026-PRESENT :
<https://github.com/SAMEER-MLPROJECT/UNIAETHEL>